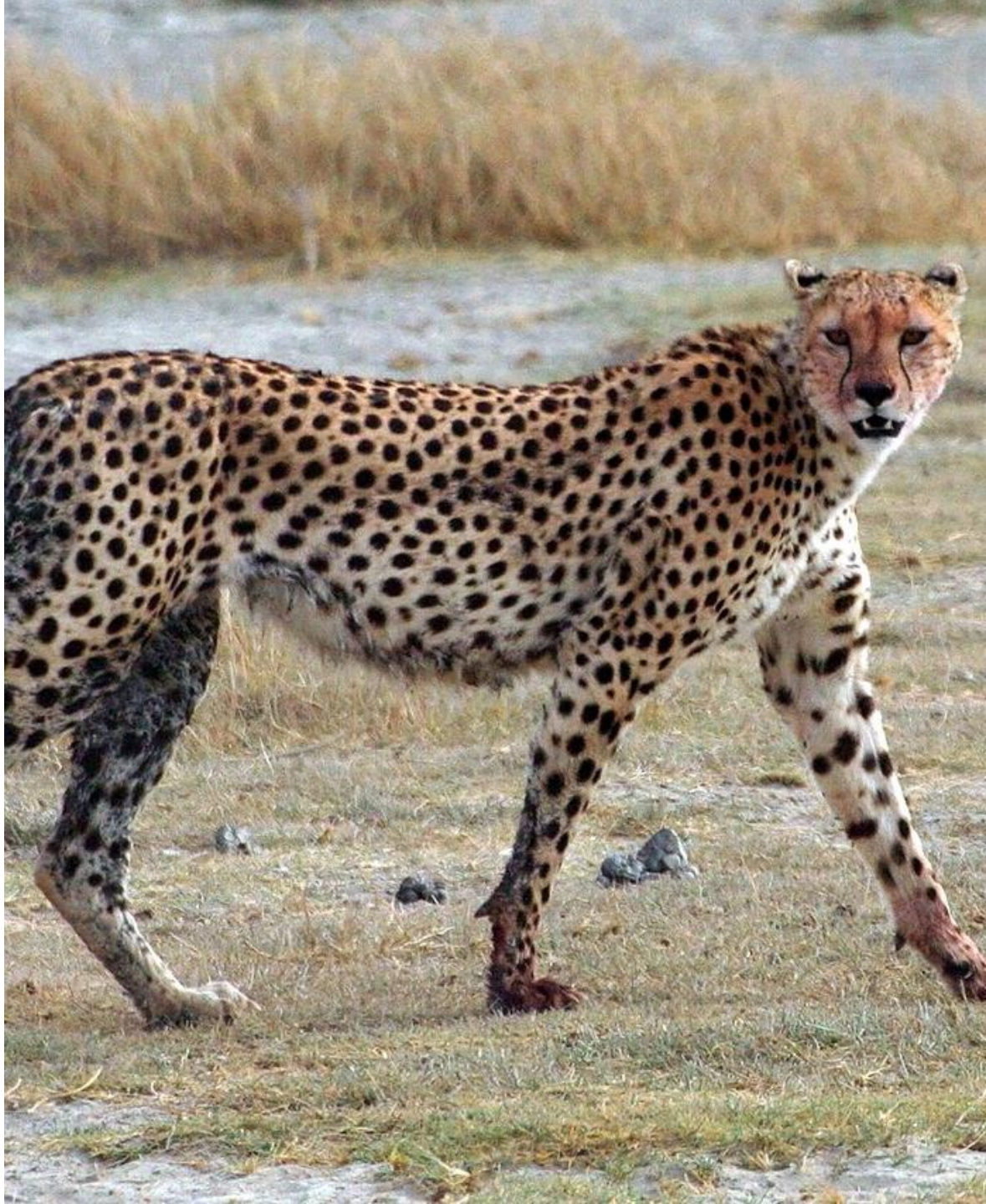
DP Biology – C3.1

# Integration of body systems

- What do nerves and hormones each do to coordinate body systems?
- How does feedback help to control body systems?

Statements C3.1.1–16, SL and HL. Textbook pp. 490–509. Plants are in Plant Systems.
Image: cheetah. Author's modification of an original by Rob, CC BY 2.0.



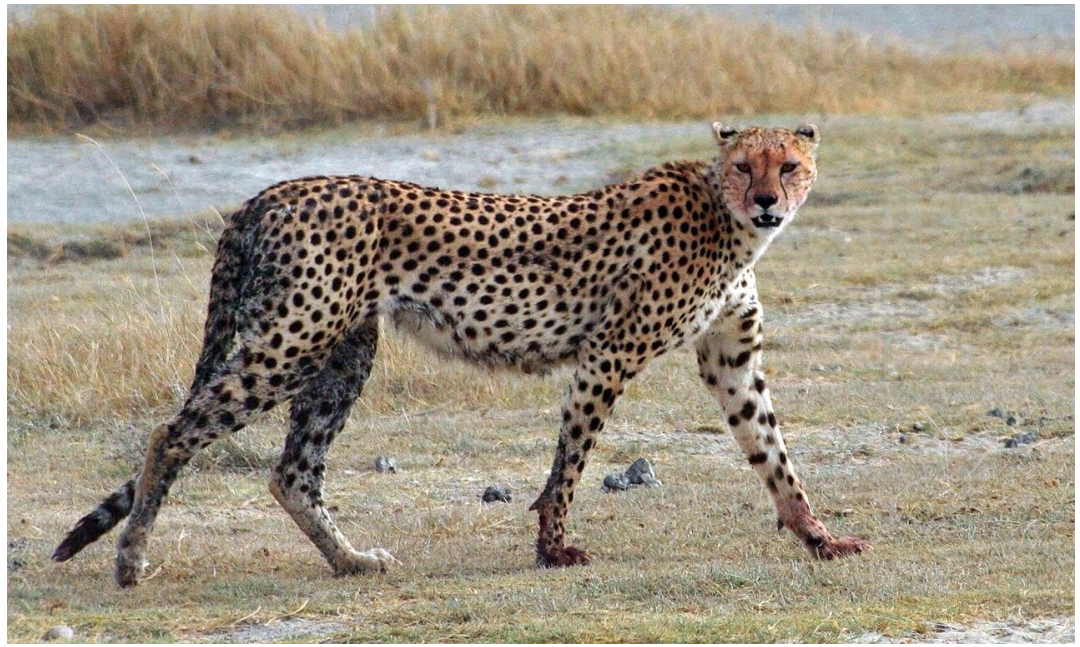
# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | C3.1.1–8 | Explain integration, compare nerves and hormones, and outline the roles of the brain, spinal cord, neurons and nerves |
| **2** | C3.1.9–13 | Describe the pain reflex arc and the cerebellum, and explain melatonin, epinephrine and the hypothalamus |
| **3** | C3.1.14–16 | Explain feedback control of heart rate and ventilation, and the control of peristalsis |

# Systems work together

C3.1.1–2

Cells → Tissues → Organs → Organ systems → **Organism**



Cheetah. Author's modification of an original by Rob, CC BY 2.0.

**Integration** is the coordination of many parts so that together they do one job.

A cheetah's fast muscle, large heart and flexible spine only make a hunter when they work together. This is an **emergent property**: the whole can do what no part can.

### Think

Running for a bus: which systems must change, and what if only your muscles sped up?

# Nervous and hormonal signalling

C3.1.3

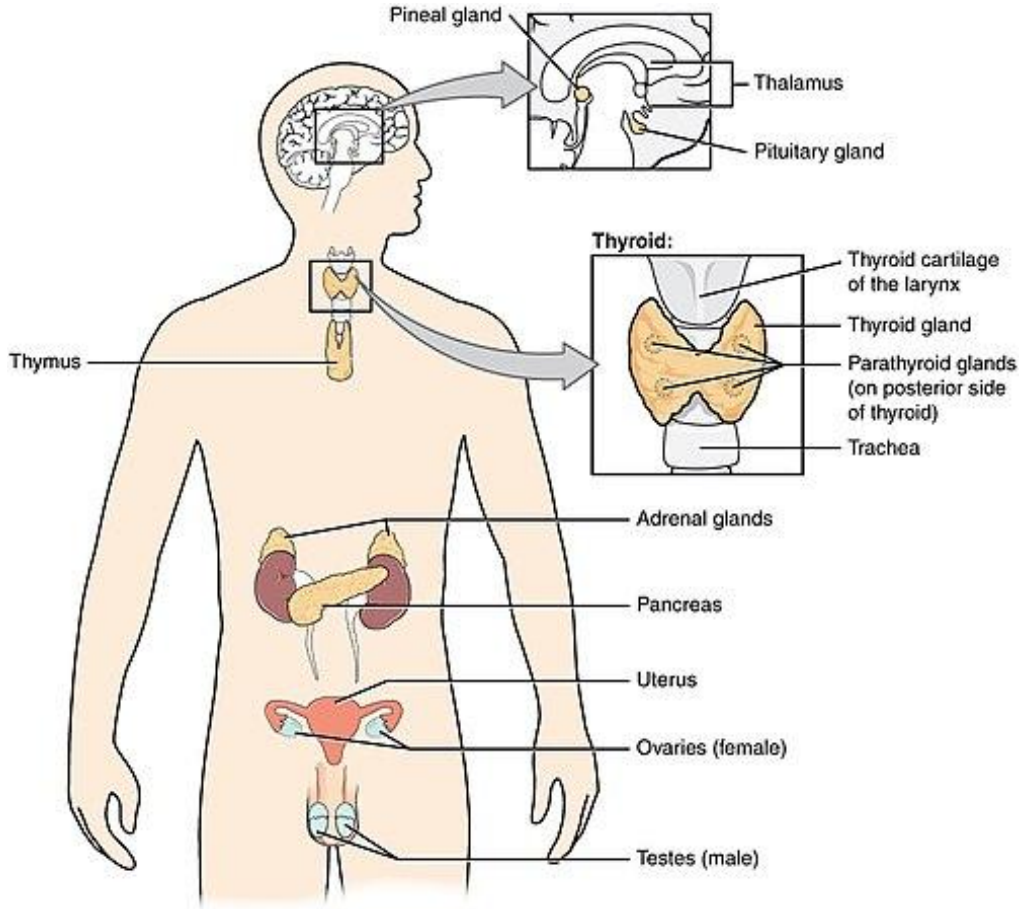
| | Nervous | Hormonal |
|---|---|---|
| **Signal** | Electrical impulses, neurotransmitters | Chemical messengers |
| **Travels in** | Neurons and nerves | The blood |
| **Reaches** | Specific muscles or glands | Target cells anywhere |
| **Speed** | Very fast | Slower |
| **Effect lasts** | Short time | Long time |

**Examples**

Nervous: blinking, the pain reflex. Hormonal: growth, reproductive cycles, homeostasis.

# The endocrine system






Endocrine glands. OpenStax College, CC BY 3.0.

**Endocrine glands** have no ducts. They secrete **hormones** straight into the blood, which carries them everywhere.

Only **target cells** with the right receptors respond. Tiny amounts are enough.

# Blood links organs by transport

C3.1.3

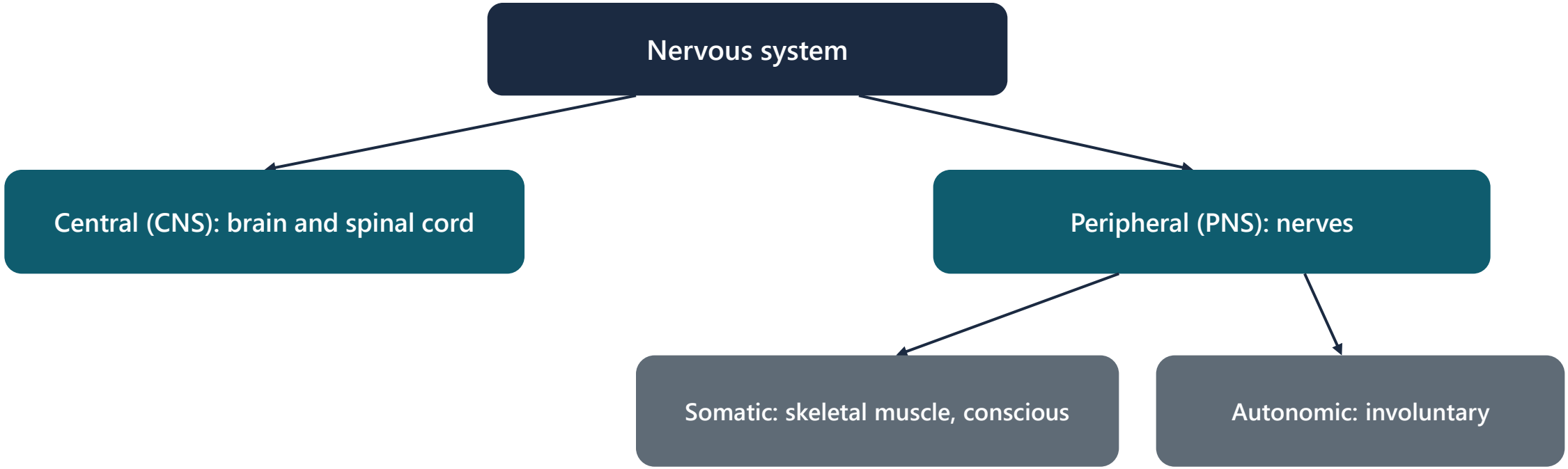
| Material | From | To |
| --- | --- | --- |
| **Glucose** | Small intestine, liver | All respiring cells |
| **Oxygen** | Lungs | All respiring cells |
| **Carbon dioxide** | Respiring cells | Lungs |
| **Urea** | Liver | Kidneys |
| **Hormones** | Endocrine glands | Target cells |

**Key idea**

Blood integrates organs by moving materials and energy-rich molecules between them.

# How the nervous system is organized

C3.1.4

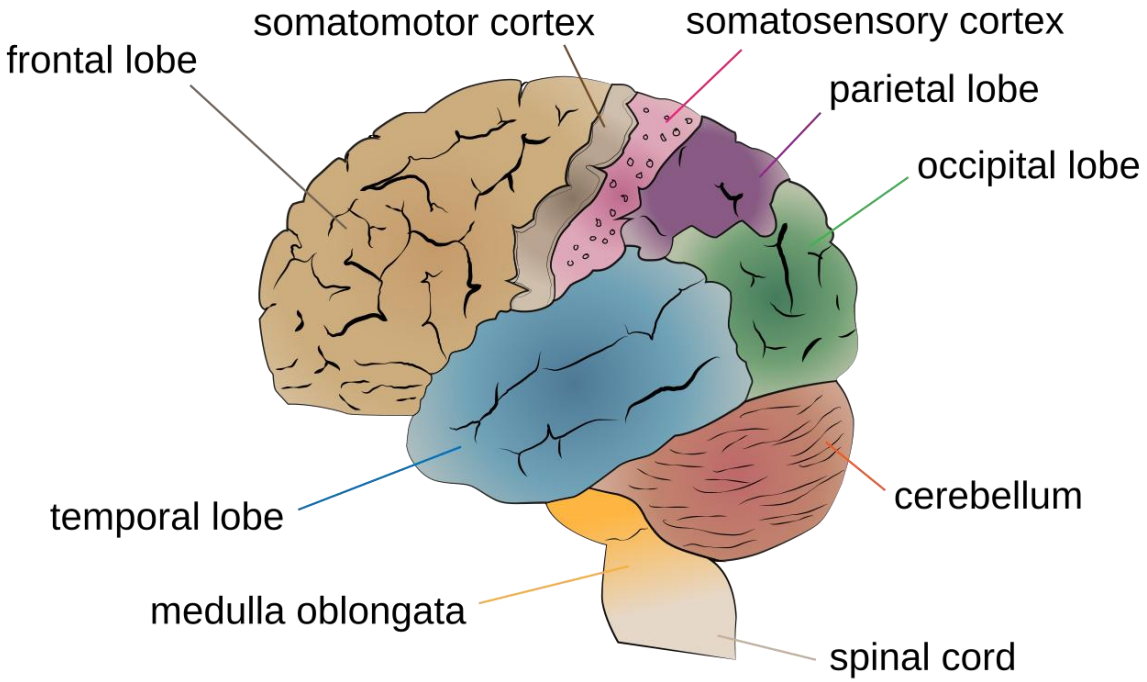



## Autonomic branches

**Sympathetic** signals prepare for action; **parasympathetic** signals relax. The **enteric** nervous system in the gut wall works on its own.

# The brain integrates information

C3.1.4




Cerebral hemisphere and its regions. Jkwchui, CC BY-SA 3.0.

The brain **combines inputs** from many receptors at once, decides, and sends impulses to muscles and glands.

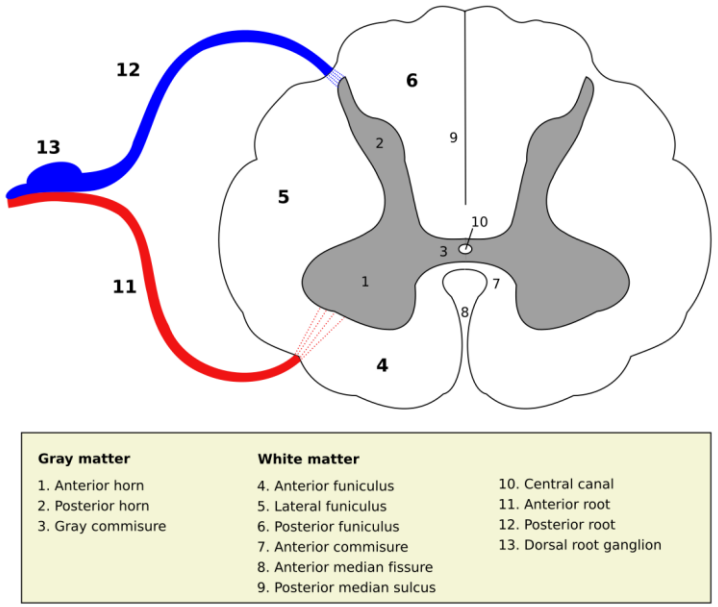
It also **stores** information (learning and memory) and starts its own activity: thought and planning.

**Exam note**

Keep this general: the brain is the main integrating centre.

# The spinal cord

C3.1.5




Spinal cord in section. Polarlys, CC BY 2.5.

Carries impulses between brain and body, and handles **unconscious** processes such as reflexes without the brain.

**Grey matter**: cell bodies and synapses. **White matter**: myelinated fibres.

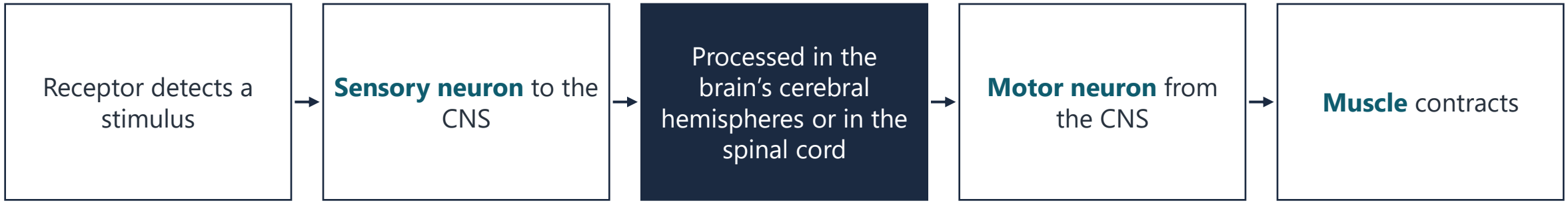
| Conscious | Unconscious |
|---|---|
| **Voluntary movement** | Heart rate |
| **Thought, speech** | Pupil size, moving food |

## Common mistake

Not all brain activity is conscious. Conscious means deliberate and aware; unconscious means automatic.

# Input and output through neurons

C3.1.6–7




**Sensory** neurons bring input into the CNS, where it is processed; **motor** neurons carry output out to the muscles.

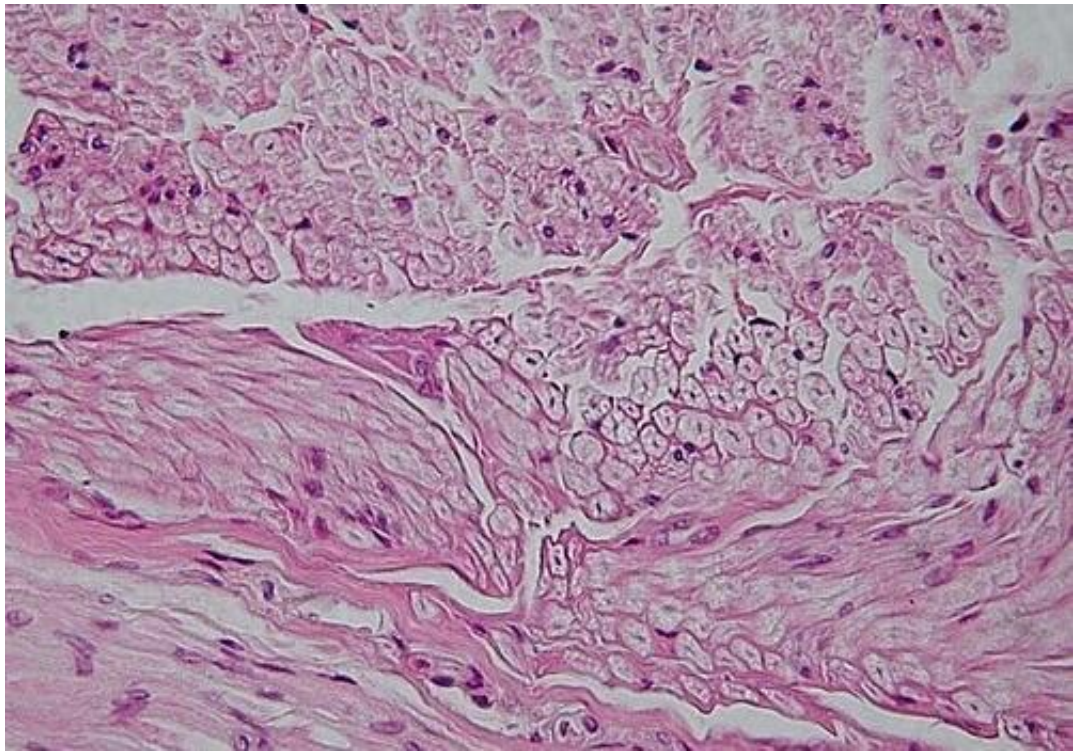
The left hemisphere controls the right side of the body, and the right controls the left.

### Example

Writing with a pen: touch receptors send input; the hemispheres plan; motor neurons move the fingers, with constant adjustment.

# Nerves are bundles of nerve fibres

C3.1.8



A peripheral nerve in section. CC BY-SA 3.0.

A **nerve** holds fibres from both **sensory and motor neurons**, grouped in bundles inside a protective sheath.

Most nerves have both **myelinated** and **unmyelinated** fibres.

**Common mistake**

A neuron is one cell. A nerve is a bundle of many fibres from many neurons.

# Practice: the falling ruler

C3.1.6–7

A student caught a dropped ruler five times. The values are invented for practice; the pattern is realistic.

| Trial | Distance fallen / cm |
|---|---|
| 1 | 18 |
| 2 | 15 |
| 3 | 14 |
| 4 | 12 |
| 5 | 12 |

1. Calculate the mean distance.
2. Reaction time t = square root of (2d / 9.8), with d in metres. Calculate t for the mean.
3. Trace the signal from eye to fingers, naming each neuron type.
4. Suggest why the distance fell with practice.

# Check-in: class 1

C3.1.1–8

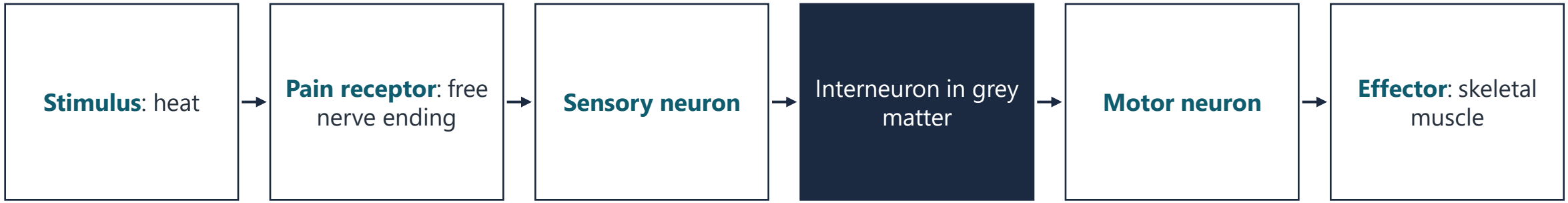
1. Define integration and give an example of an emergent property.
2. Compare nervous and hormonal signalling in three ways.
3. State what the CNS is made of.
4. Distinguish between grey and white matter.
5. Distinguish between a neuron and a nerve.

**How to use this**

Answer without your notes, then check against slides 3–12.

# The pain reflex arc

C3.1.9




A **reflex** is a rapid, **involuntary** response to a stimulus. The hand moves before you feel the pain, because the signal does not need to reach the brain first.

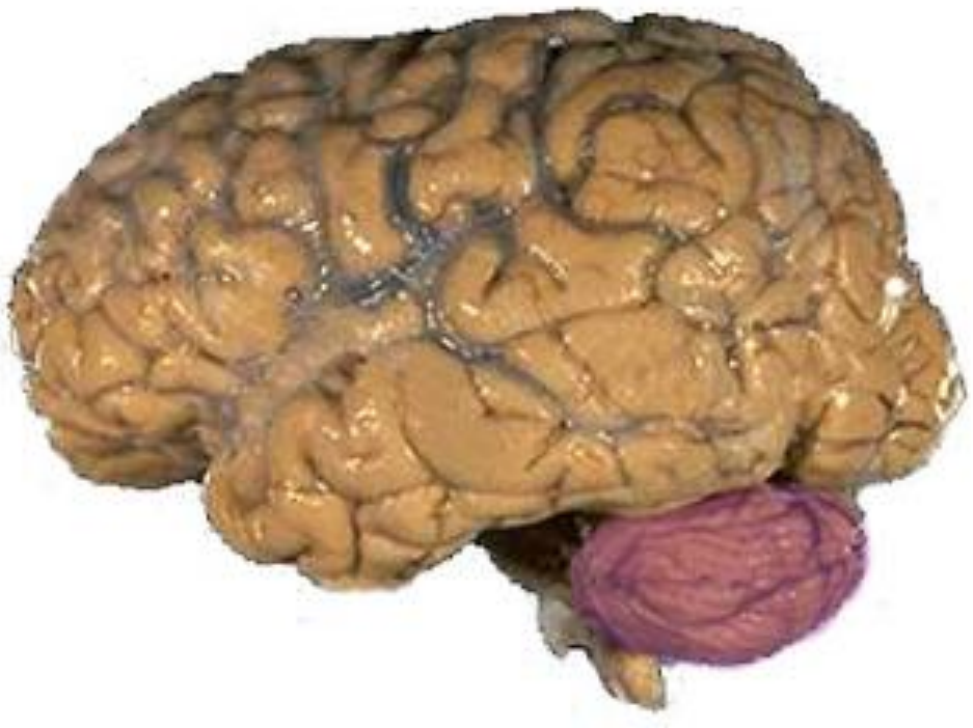
Other fibres do carry the message to the brain, so you become aware of it a moment later.

### Exam note

Learn this example: free nerve ending in the hand, one interneuron in the grey matter of the spinal cord, skeletal muscle as the effector.

# The cerebellum

C3.1.10



The cerebellum. NIH, public domain.

At the back of the brain. It **coordinates** skeletal muscle contraction, making movements smooth and precise, and controls **balance** and posture.

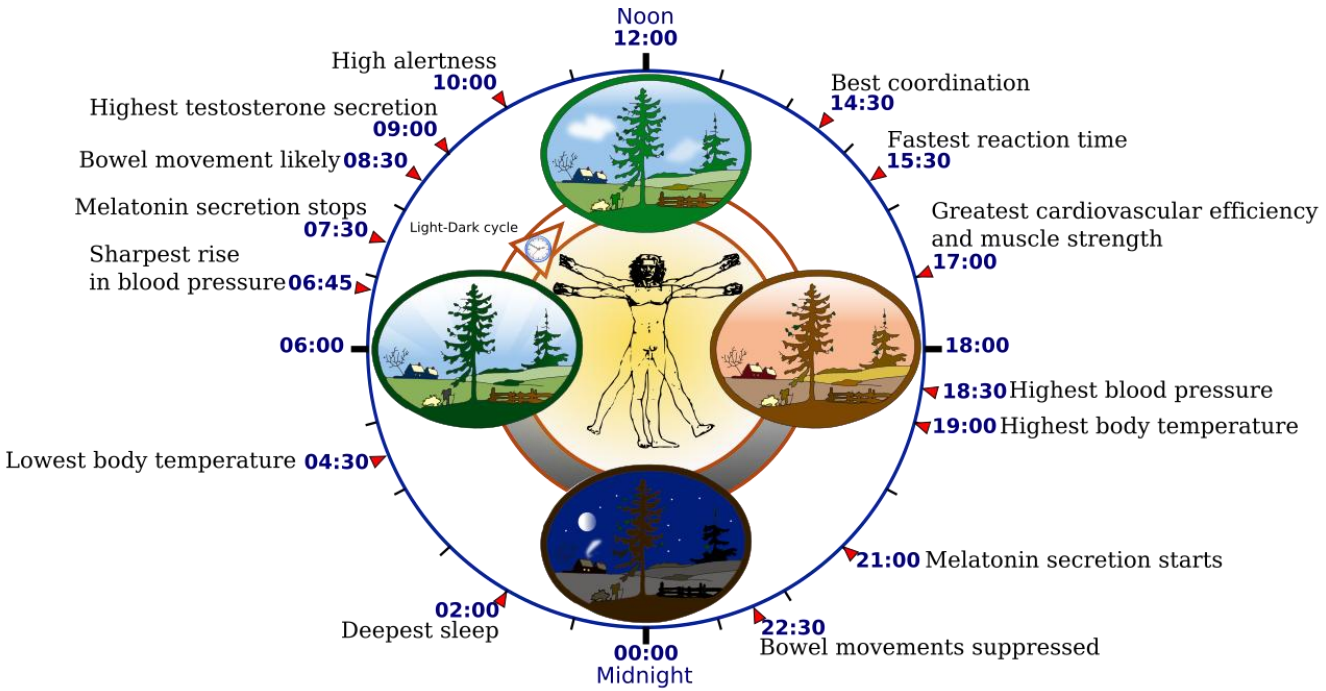
It coordinates movement but does not start it.

### Think

Alcohol affects the cerebellum. Which roadside tests for drunk drivers would show this?

# Melatonin and circadian rhythms

C3.1.11




The human circadian rhythm. YassineMrabet, CC BY-SA 3.0.

**Circadian rhythms** follow a cycle of about 24 hours.

The **pineal gland** secretes **melatonin** in the dark. High melatonin promotes sleep; falling levels prepare the body for waking.

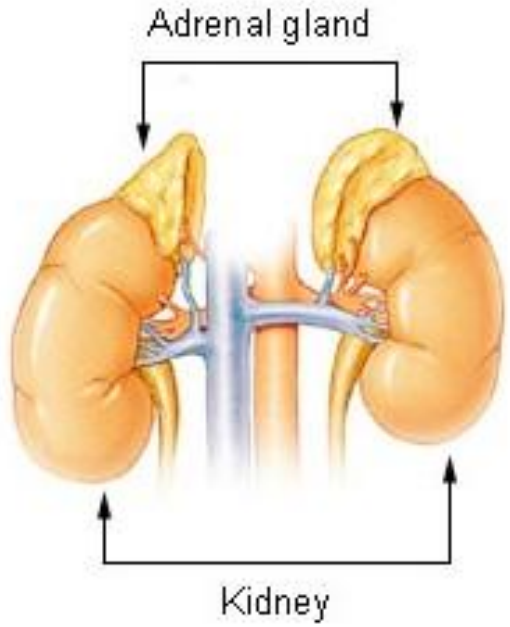
## Think

Why can jet lag last several days?

# Epinephrine: ready for action

C3.1.12

| Effect | Why it helps |
|---|---|
| **Heart rate rises** | More oxygen and glucose to muscles |
| **Breathing rate rises** | More $O_2$ in, more $CO_2$ out |
| **Liver glycogen broken down** | Blood glucose rises |
| **Blood diverted to muscles** | Supply goes where it is needed |



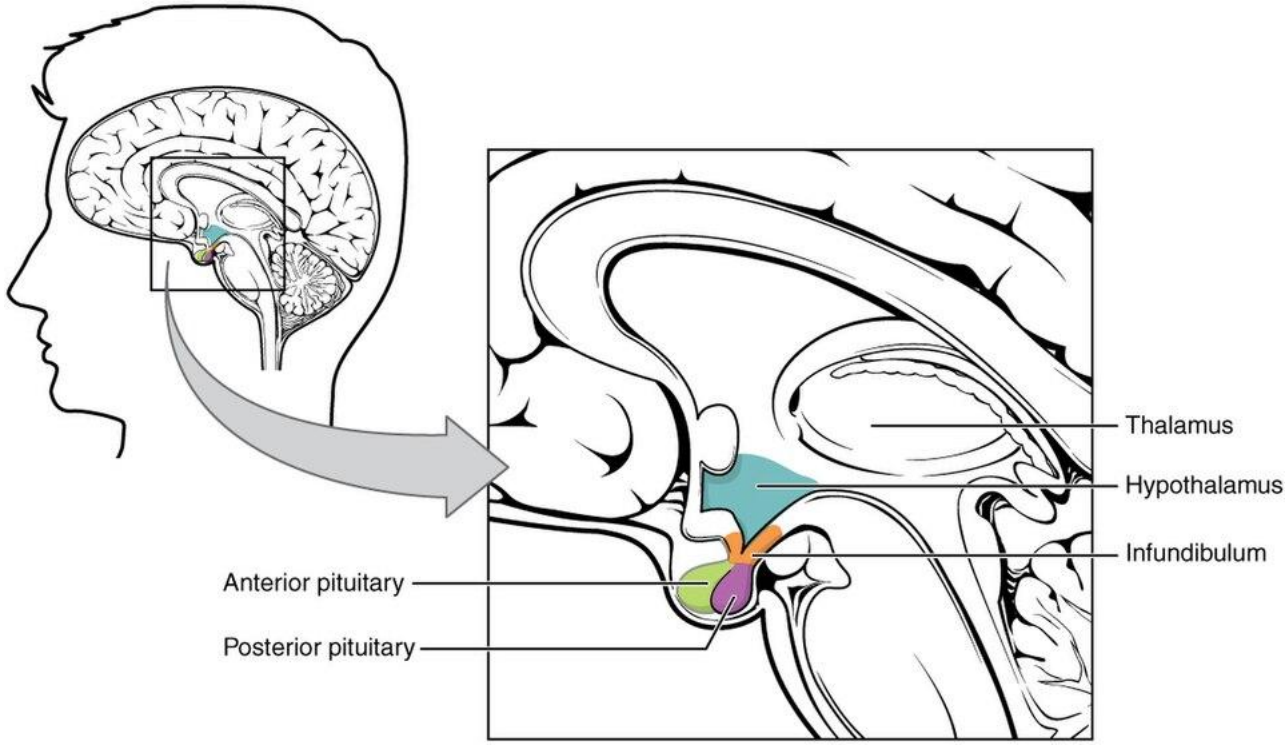

Adrenal gland on the kidney. Public domain.

## Key idea

The **adrenal glands** secrete **epinephrine** (adrenaline) under stress. It travels in the blood and readies the whole body for vigorous activity: fight or flight.

# The hypothalamus and pituitary

C3.1.13




The hypothalamus and pituitary gland. OpenStax College, CC BY 3.0.

The **hypothalamus** links the nervous and endocrine systems. It monitors the blood and controls the **pituitary gland**, by hormones and nerve impulses.

Pituitary hormones direct other glands: growth, reproduction and homeostasis.

**Exam note**

A general understanding is enough; details of the two pituitary lobes are not needed.

# Working with data: melatonin over a day

C3.1.11

Blood melatonin of one adult. The values are invented for practice; the pattern is realistic.

| Time of day | Melatonin / pg $ml^{-1}$ |
|---|---|
| **18:00** | 10 |
| **21:00** | 30 |
| **00:00** | 70 |
| **03:00** | 85 |
| **06:00** | 45 |
| **09:00** | 8 |

1. Describe the pattern.
2. State when the person is likely to feel sleepiest.
3. Predict the effect of a bright screen at 23:00.
4. Explain why melatonin falls after 06:00.

# Check-in: class 2

C3.1.9–13

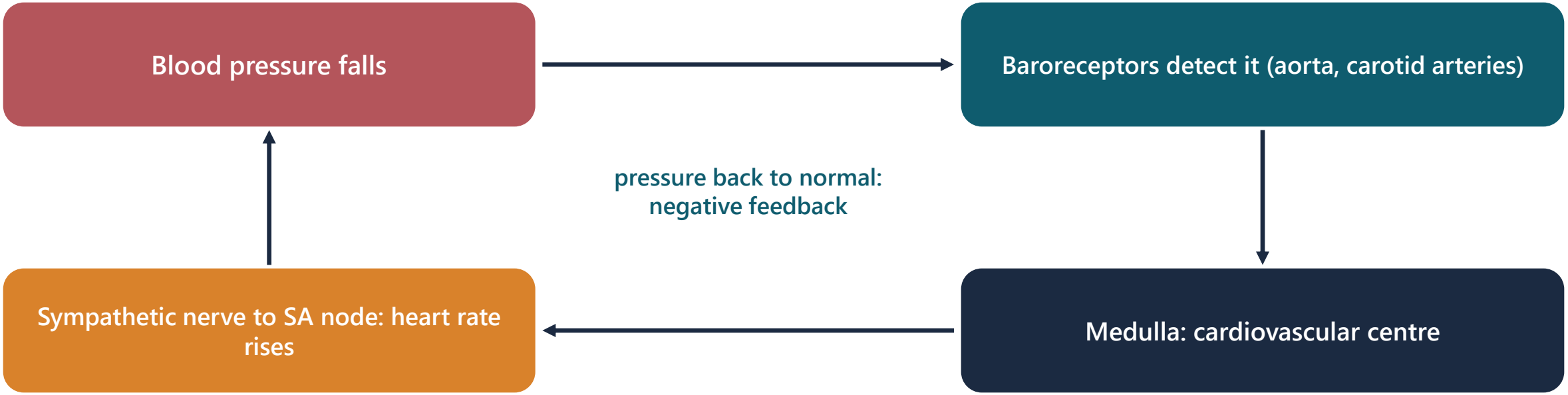
1. List the parts of the pain reflex arc in order.
2. State two roles of the cerebellum.
3. Explain how melatonin controls sleep.
4. Outline three effects of epinephrine.
5. Describe the role of the hypothalamus.

**How to use this**

Answer without your notes, then check against slides 14–19.

# Feedback control of heart rate

C3.1.14




**Chemoreceptors** (in the carotid arteries, aorta and brainstem) detect high $CO_2$ and low pH, which also raises heart rate.

The **vagus nerve** slows the heart when pressure is high. The SA node sets the beat; nerves only change its rate.

# Speeding up and slowing down

C3.1.14

| Change detected | Nerve | Heart rate |
|---|---|---|
| **Low blood pressure** | Sympathetic | Rises |
| **High $CO_2$, low pH, low $O_2$** | Sympathetic | Rises |
| **High blood pressure** | Vagus (parasympathetic) | Falls |
| **Low $CO_2$, higher pH** | Vagus (parasympathetic) | Falls |

**Key idea**

In exercise, a faster heart carries $CO_2$ to the lungs sooner, so blood pH returns to safe limits.

# Feedback control of ventilation rate

C3.1.15




The respiratory centre in the brainstem controls breathing automatically. After exercise, $CO_2$ falls and so does the ventilation rate.

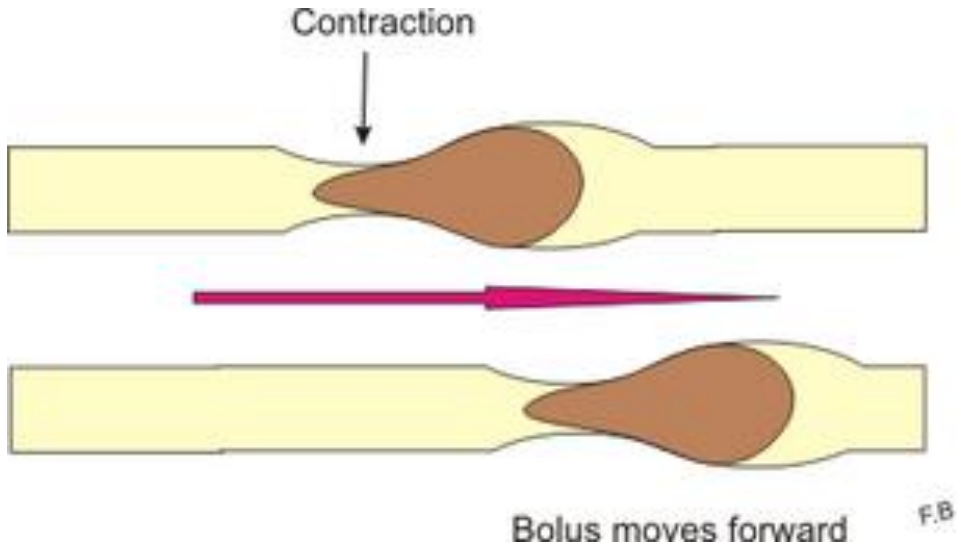
You can override it for a while, as when you speak or sing.

**Common mistake**

The main trigger to breathe faster is a rise in $CO_2$, detected as a fall in pH, not usually a lack of oxygen.

# Control of peristalsis

C3.1.16




Peristalsis. Boumphreyfr, CC BY-SA 3.0.

**Peristalsis** is waves of muscle contraction along the gut wall. Circular and longitudinal muscles act in turn.

It is involuntary, coordinated by the **enteric nervous system** in the gut wall.

| Stage | Control | System |
|---|---|---|
| **Swallowing; egesting faeces** | Voluntary | Central nervous system |
| **Oesophagus to rectum** | Involuntary | Enteric nervous system |

# Working with data: heart and breathing in exercise

C3.1.14–15

One student before, during and after 4 minutes of stepping. The values are invented for practice; the pattern is realistic.

| Time / min | Heart rate / $min^{-1}$ | Breaths / $min^{-1}$ |
|---|---|---|
| **0 (rest)** | 68 | 14 |
| **2 (exercise)** | 118 | 24 |
| **4 (exercise)** | 132 | 28 |
| **6 (recovery)** | 96 | 19 |
| **8 (recovery)** | 74 | 15 |

1. Calculate the percentage increase in heart rate from rest to 4 minutes.
2. Explain how chemoreceptors raise the ventilation rate.
3. Explain why both rates are still raised at 6 minutes.
4. Suggest why an athlete’s heart rate would rise less.

# Check-in: class 3

C3.1.14–16

1. State what baroreceptors and chemoreceptors detect.
2. Explain how heart rate rises when blood pressure falls.
3. Explain why ventilation rate rises in exercise.
4. Explain why these are examples of negative feedback.
5. Distinguish between the control of swallowing and of peristalsis.

**How to use this**

Answer without your notes, then check against slides 21–25.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Reflexes are controlled by the brain.** | Pain reflexes are processed in the spinal cord. |
| **Hormones travel along nerves.** | Hormones travel in the blood. |
| **Nerves make the heart beat.** | The SA node sets the beat; nerves change its rate. |
| **Ventilation is controlled mainly by oxygen.** | Mainly by $CO_2$, detected as a fall in pH. |
| **A nerve is a single neuron.** | A nerve is a bundle of many fibres. |

# Summary: integration of body systems

C3.1.1–16

| | Key idea |
|---|---|
| **C3.1.1–3** | Parts coordinated by nerves, hormones and blood |
| **C3.1.4–8** | Brain integrates; spinal cord for unconscious control; nerves hold sensory and motor fibres |
| **C3.1.9–10** | Pain reflex through the spinal cord; cerebellum for coordination and balance |
| **C3.1.11–13** | Melatonin for sleep; epinephrine for action; hypothalamus controls the pituitary |
| **C3.1.14–15** | Receptors, medulla and nerves adjust heart and breathing rates |
| **C3.1.16** | Enteric nervous system controls peristalsis |

# Check that you can...

1. **Compare** nervous and hormonal signalling
2. **Outline** the roles of the brain, spinal cord and nerves
3. **Describe** a pain reflex arc
4. **State** the roles of the cerebellum, melatonin and epinephrine
5. **Outline** control by the hypothalamus and pituitary
6. **Explain** feedback control of heart rate and ventilation rate
7. **Outline** the control of peristalsis

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. Give two differences between nervous and hormonal signalling.
2. List, in order, the parts of a pain reflex arc from receptor to effector.
3. Explain why ventilation rate rises during exercise, starting with respiration in muscle.

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